

Hemal Shah

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Professional Summary

Backend-focused Computer Science engineer passionate about building scalable real-world systems, automation infrastructure, and AI-powered backend applications. Experienced in FastAPI, PostgreSQL, Docker, APIs, workflow orchestration, and production deployments. Strong interest in startup environments, product architecture, backend scalability, and solving practical engineering problems from scratch.

Technical Skills

- **Languages:** Python, JavaScript, SQL, C++, C#
- **Backend:** FastAPI, REST APIs, JWT Authentication, MCP Servers
- **Database:** PostgreSQL, MongoDB, Supabase, pgvector
- **DevOps/Cloud:** Docker, Docker Compose, Railway, Render, CI/CD, Linux
- **Automation & AI:** n8n, OCR Pipelines, RAG, GraphRAG, API Integrations
- **Other:** Git, GitHub, Web Scraping, Async Systems, Workflow Automation

Experience

Python & Automation Developer Intern — Inddig Media Pvt Ltd

Mar 2026 – Present

- Architected and deployed backend infrastructure for AI-driven medical appointment and scheduling systems using FastAPI and PostgreSQL.
- Integrated Google Calendar APIs, OAuth verification systems, DNS routing, and production hosting pipelines for enterprise-grade applications.
- Built scalable workflow automation systems using n8n for 400+ client profiles, reducing manual operational overhead significantly.
- Managed deployment pipelines, backend integrations, hosting infrastructure, SSL setup, and real-time production updates.

AI Engineer Intern — SentimentAI

Jul 2025 – Jul 2025

- Developed backend automation systems for OCR-based invoice processing, intelligent document analysis, and AI-powered workflows.
- Built scalable scrapers for Google Maps and LinkedIn using Python, BeautifulSoup, Puppeteer, and API-driven automation pipelines.
- Created Telegram AI systems integrated with n8n for conversational document processing and workflow orchestration.
- Worked extensively with asynchronous APIs, real-time automation workflows, intelligent data extraction, and backend integrations.

Projects

Business Scraper MCP

- Built a production-ready MCP (Model Context Protocol) backend server for scraping and exporting business leads from Google Maps using FastMCP, httpx, pandas, and Prefab UI.
- Implemented asynchronous business search systems, Google Places API integrations, Apify fallback architecture, pagination handling, and concurrent email extraction pipelines.
- Designed modular backend architecture with Pydantic validation, Docker deployment, rate limiting, structured logging, CSV export systems, and scalable API tooling.

TrendGraphAI

- Developed an AI-powered research intelligence platform using FastAPI, PostgreSQL, pgvector, Docker, and GraphRAG architecture.
- Implemented semantic vector search, automated research ingestion pipelines, intelligent retrieval systems, and scalable backend APIs.
- Integrated AI models and embedding systems using Sentence Transformers and Groq APIs for intelligent document analysis workflows.

GitLite

- Built a GitHub-style backend application using FastAPI with JWT authentication, Docker containerization, and production deployment on Render.
- Developed secure REST APIs, authentication systems, protected routes, and scalable backend workflows for repository-style operations.

Telegram AI Bot

- Developed an AI-powered Telegram automation bot using n8n and external APIs for PDF/Image analysis and intelligent conversational workflows.
- Integrated OCR pipelines, HTTP request automation, and asynchronous workflow orchestration without traditional back-end dependencies.

Education

GSFC University

2022 – 2026

B.Tech in Computer Science & Engineering

CGPA: 7.12 / 10